HOMEWORK 5

1.) $x_t + 1.7x_{t-1} + 0.6x_{t-2} = \varepsilon_t - 1.4\varepsilon_{t-1} + 0.4\varepsilon_{t-2}$

$a(z) = 1 + 1.7z + 0.6z^2$

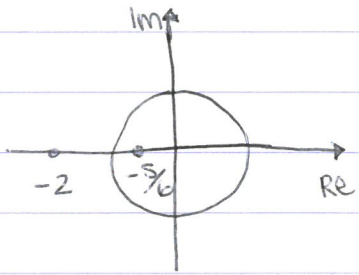
$\Rightarrow 1 + 1.7z + 0.6z^2 = 0$

$\Rightarrow 6z^2 + 17z + 10 = 0$

Using the roots equation

$$z = \frac{-17 \pm \sqrt{17^2 - 4 \cdot 6 \cdot 10}}{12}$$

$= -5/6, -2$



✓ As one of the roots lies within the complex unit circle the process is not causal.

✓ Also as neither of the roots lies on the complex unit circle the process is stationary.

2.) $x_t + 1.2x_{t-1} + 0.6x_{t-2} = \varepsilon_t - 1.4\varepsilon_{t-1} + 0.5\varepsilon_{t-2}$

$a(z) = 1 + 1.2z + 0.6z^2$

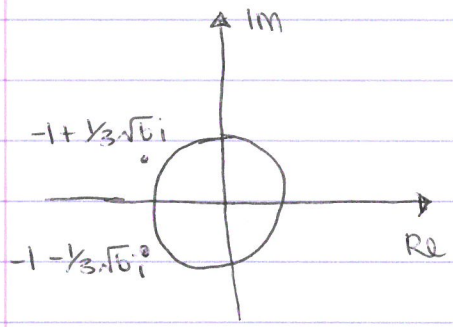
$\Rightarrow 1 + 0.6z^2 + 1.2z = 0$

$\Rightarrow 6z^2 + 12z + 10 = 0$

Using the roots equation

$$z = \frac{-12 \pm \sqrt{12^2 - 4 \cdot 6 \cdot 10}}{12} = \frac{-12 \pm \sqrt{-96}}{12} = \frac{-12 \pm i\sqrt{96}}{12}$$

$$= \frac{-1 \pm i\sqrt{16 \cdot 6}}{12} = -1 \pm \frac{4}{12} i\sqrt{6} = -1 \pm \frac{1}{3} i\sqrt{6}$$



The roots do not lie within the complex unit circle so the process is causal and as the roots do not lies on the complex unit circle, the process is stationary.

3.) $X_t + 1.3X_{t-1} + 0.3X_{t-2} = \varepsilon_t - 1.4\varepsilon_{t-1} + 0.6\varepsilon_{t-2}$

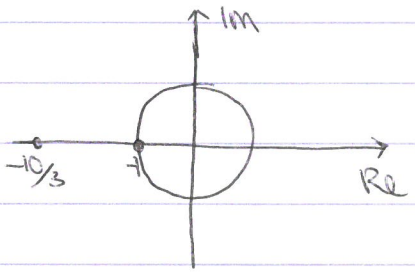
$a(z) = 1 + 1.3z + 0.3z^2$

$\Rightarrow 0.3z^2 + 1.3z + 1 = 0$

$\Rightarrow 3z^2 + 13z + 10 = 0$

$\Rightarrow (3z+10)(z+1) = 0$

so $z = -10/3$, $z = -1$



As one of the roots does lie within the complex unit circle the process is not causal, but as the root $z=-1$ lies on the complex unit circle the process is not stationary.

4) $X_t - 0.3X_{t-1} - 0.7X_{t-2} = \varepsilon_t + 0.9\varepsilon_{t-1} - 0.5\varepsilon_{t-2}$

$b(z) = 1 + 0.9z - 0.5z^2$

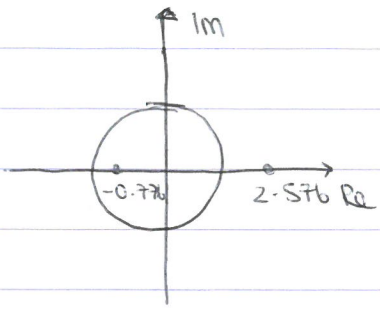
$\Rightarrow 0.5z^2 - 0.9z - 1 = 0$

$\Rightarrow 5z^2 - 9z - 10 = 0$

Using the roots equation

$$z = \frac{9 \pm \sqrt{81 - 4(-50)}}{10} = \frac{9 \pm \sqrt{281}}{10}$$

$$z = -0.776,\ 2.576$$



As the root $z = -0.776$ lies within the complex unit circle the process is not invertible ✓

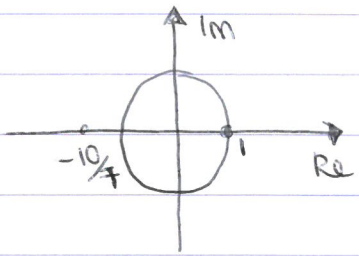
Then $a(z) = 1 - 0.3z - 0.7z^2$

$$\Rightarrow \quad 7z^2 + 3z - 10 = 0$$

Using the roots equation

$$z = \frac{-3 \pm \sqrt{9 + 4 \cdot 7 \cdot 10}}{14}$$

$$= -10/7,\ 1$$



As the root of $a(z)$, $z = 1$, lies on the complex unit circle the process is not stationary. ✓

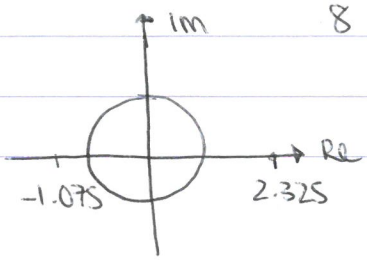
5.) $x_t - 0.3x_{t-1} + 0.7x_{t-2} = \varepsilon_t + 0.5\varepsilon_{t-1} - 0.4\varepsilon_{t-2}$

$b(z) = 1 + 0.5z - 0.4z^2$

$$\Rightarrow \quad 4z^2 - 5z - 10 = 0$$

Using the roots equation

$$z = \frac{5 \pm \sqrt{25 + 4 \cdot 4 \cdot 10}}{8} = \frac{5 \pm \sqrt{185}}{8} = -1.075,\ 2.325$$



✓ As the roots of $b(z)$ do not lie within the complex unit circle the process is invertible.

$$a(z) = 1 - 0.3z + 0.7z^2$$
$$\rightarrow \quad 7z^2 - 3z + 10 = 0$$

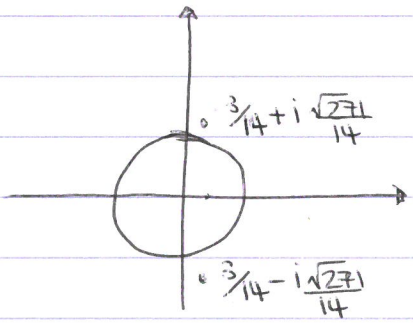
Using the roots equation

$$z = \frac{3 \pm \sqrt{9 - 4 \cdot 7 \cdot 10}}{14}$$

$$= \frac{3 \pm \sqrt{-271}}{14}$$

$$= \frac{3 \pm i\sqrt{271}}{14}$$

$$= \frac{3}{14} \pm i\frac{\sqrt{271}}{14}$$



✓ As the roots of $a(z)$ do not lie on the complex unit circle the process is stationary.